#ManyDaughters - Pre-Analysis-Plan

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General remarks

The hypotheses leave room for interpretation. It is not clear whether one should focus on people with children only and among those with children compare those with daughters to those without daughters. It might also be interpreted as having daughters at all, compared to not having children and those having only sons. Similarly, one might focus on parents that exclusively have daughters or exclusively have sons.

We decided to compare people with at least one daughter to parents without daughters.

Exclusion / Inclusion Criteria

We only use respondents from private households and exclude participants below 18 years of age.

To determine age at the time of taking the survey we use *syear* - *gebjahr*, i.e., the survey year minus the birth year and check whether the difference is smaller than 18. If so, we exclude the respondent.

We exclude everyone who is not a respondent from a private household by conditioning on *pop* being 1 or 2.

Main explanatory variable

We construct indicators for having daughters. The basis for the indicator is variable pl/pld0026 (number of daughters). Until the first time a respondent encounters the question, we code the indicator variable as missing. If they indicate “not applicable”, we treat this as having kids, but not a daughter. We assume it is a son and code has_daughters = 0. If the number of daughters is positive, we code has_daughters = 1. The question is not asked at every survey participation. Therefore, we forward fill the values across years. We assume that once a participant has a daughter, they always have a daughter for the purpose of our study. We forward-fill both zeros (having at last one son) and ones (having at least one daughter).

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).

Remarks

There is room for interpretation regarding the "support for gender equality". This may be interpreted as gender equality in the family when it comes to responsibilities. It could also be interpreted more generally in the sense of equality between men and women in society (including income, etc.). It could also be interpreted as gender equality in general, not limited to the classical roles of men and women.

Dependent variable

We operationalize this by using question *plh0308_v2*, that reads "Am besten ist es, wenn der Mann und die Frau beide gleich viel erwerbstätig sind und sich beide in gleichem Maße um Haushalt und Familie kümmern". We exclude respondents whose response is coded 0 or below. Low values indicate disagreement with the statement, higher values indicate agreement with the statement.

For the analysis, we take the first year in which both the number of daughters variable and the dependent variable is available. That is, we do not exploit the panel structure and only consider one observation per individual.

Regression model

Standard OLS with robust standard errors

h1_dep_var = beta0 + beta1 * has_daughers + epsilon

Interpretation

If beta1 is significantly different from zero, we observe evidence of having daughters playing a role. If the coefficient is positive, we see this as support for the hypothesis.

H2. Having daughters leads to more left-leaning political preferences.

That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

Dependent variable

We use pl/plh0004 (political attitude) as the basis for our dependent variable to test this hypothesis. We first remove all observations with negative responses, as these indicate invalid observations. The variable is coded 0 for "all the way left" to 10 "all the way right" in integer increments. We reverse the coding to have positive coefficients in the regression imply more left leaning preferences. That is, we use:

$$h2_dep_var \;=\; 10 \;-\; pl/plh0004$$

Regression model

Standard OLS with robust standard errors

h2_dep_var = beta0 + beta1 * has_daughers + epsilon

Interpretation

If beta1 is significantly different from zero, we observe evidence of having daughters playing a role. If the coefficient is positive, we see this as support for the hypothesis.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).

Dependent variable

pl/pli0044_h (hours weekday childcare)

The datasets allow us to identify individuals and their partners. We build pairs of respondents and calculate the absolute difference in the hours spent on the variable of interest. We regress this absolute difference on the indicator variable for having daughters. As the variables indicate hours, we only take values greater or equal 0 into account. Because we build a dataset of partner pairs, we only consider pairs where both partners either indicate to have or not to have at least one daughter. It is unclear how one would treat a pair where one partner has a daughter while the other does not.

h3_dep_var = pli0044_h if the pli0044_h >= 0

The dataset is particularly rich in pli0044_h observations so we can make use of the panel structure.

Regression model

Random Effects GLS regression with standard errors clustered on the pair level.

h3_dep_var = beta0 + beta1 * has_daughers + epsilon

Interpretation

Coefficient beta1 indicates the effect of having daughters. If it is significantly different from zero and negative, we interpret this as evidence in favor of the hypothesis. In this case, having daughters would reduce the difference between the two partners in their time contribution to household work, indicating greater equality.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality, (X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

Dependent variable

The gender variable in the dataset is *sex* (1 = male, 2 = female). We use the same dependent variable as for testing h1 and implement a regression with an interaction term between the gender variable and the has_daughters variable. We define male = 1 - (sex - 1).

h4_dep_var = *plh0308_v2*

Regression model

Standard OLS with robust standard errors

h4_dep_var = beta0 + beta1 * has_daughers + beta2 * male + beta3 * (has_daughters * male) + epsilon

Interpretation

If beta3 is significantly different from zero we interpret this as evidence for the interaction of gender and having daughters to play a role. In our specification, the coefficient beta3 expresses how much stronger the effect of having daughters is for men compared to women. A positive coefficient is in line with the hypothesis.